Smart Manufacturing & Predictive Maintenance

Power BI dashboard on a SQL Server star schema

Graduation Project - Digital Egypt Pioneers Initiative (DEPI)

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Problem and objectives

- Factories record thousands of sensor readings, but teams still lack one trusted view of machine health, failures, downtime and maintenance effort.
- Goal: turn a flat 72-column file into a clean data model and a 4-page Power BI report for maintenance and plant managers.
- Success = correct numbers (reconciled with the source), clear pages, and honest limits.

100,000	5,000	10	1.47%
readings	machines	factories	readings with a failure

The data

- Kaggle: Smart Manufacturing & Predictive Maintenance (synthetic IIoT data).
- 100,000 rows x 72 columns; 5,000 machines with 20 readings each; Jan - Mar 2026.
- Quality: 0 missing values, 0 duplicate rows (verified).
- Groups: machine metadata, sensors, operating conditions, environment, maintenance history, component health, anomaly scores, targets.

Found while profiling

- maintenance_priority = 'Low' in every row
- 6 failure types (+ No Failure), not 8
- Only 1.47% of readings are failures

Approach

1 Profile

Check quality, constants per machine, label balance

2 Model

Star schema: 1 fact + 6 dimensions

3 ETL

Python splits the file and validates totals

4 Warehouse

SQL Server DDL with PK/FK, bulk load

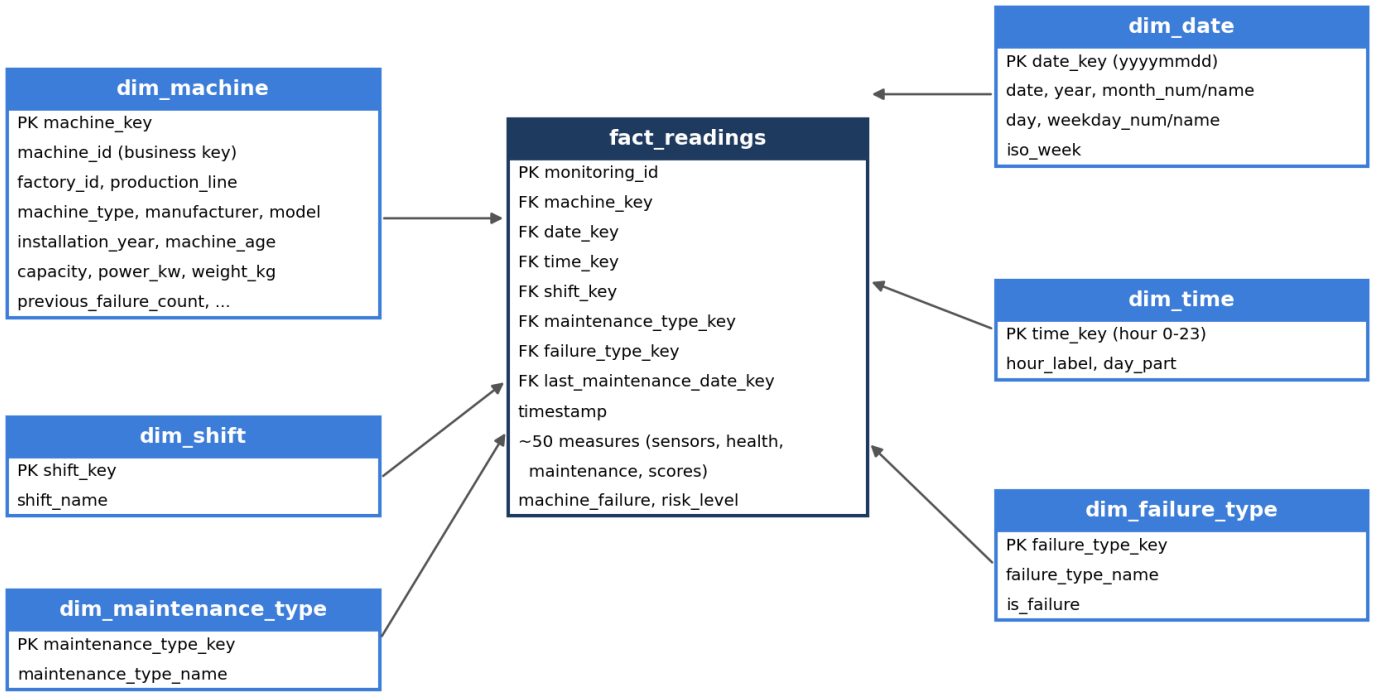
5 Report

Power BI model, DAX, 4 pages

Tools: Python (pandas), SQL Server (T-SQL), Power BI Desktop + DAX, GitHub.

Star schema design

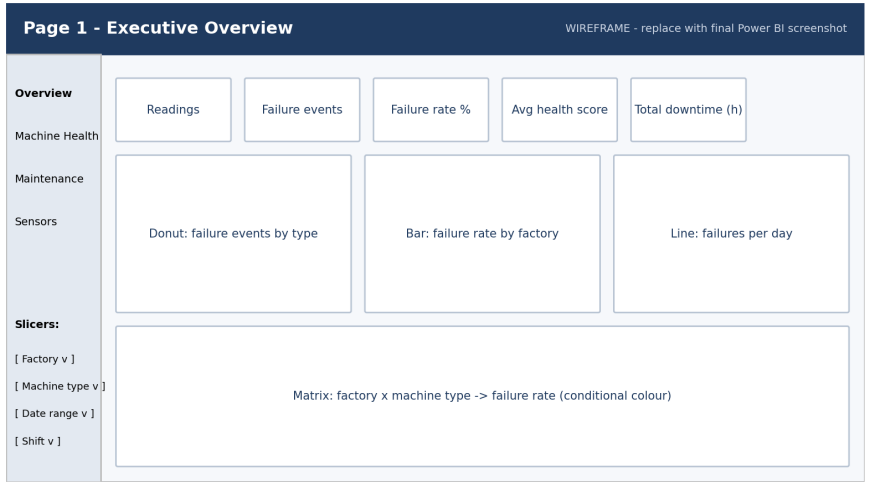
Star schema: Smart Manufacturing Data Warehouse



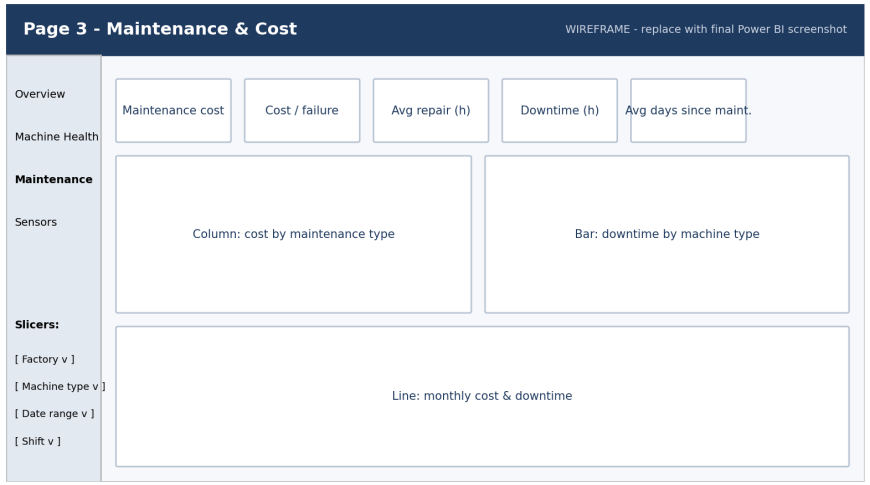
Arrows point from dimension (1) to fact (many). dim_date is also used for last_maintenance_date_key (role-playing).

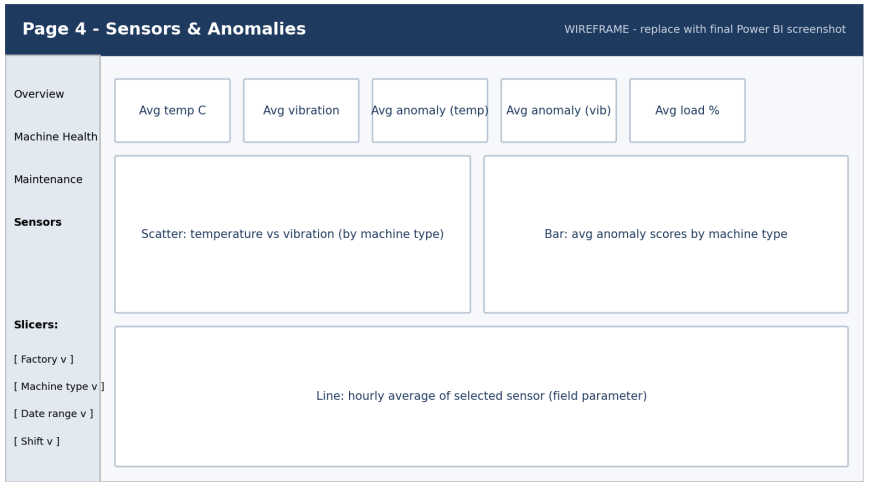
- Grain: one reading of one machine
- Surrogate keys, low-cardinality text moved to dimensions
- Checks: rows, FKs, failures and downtime totals match the source

Dashboard design (wireframes)



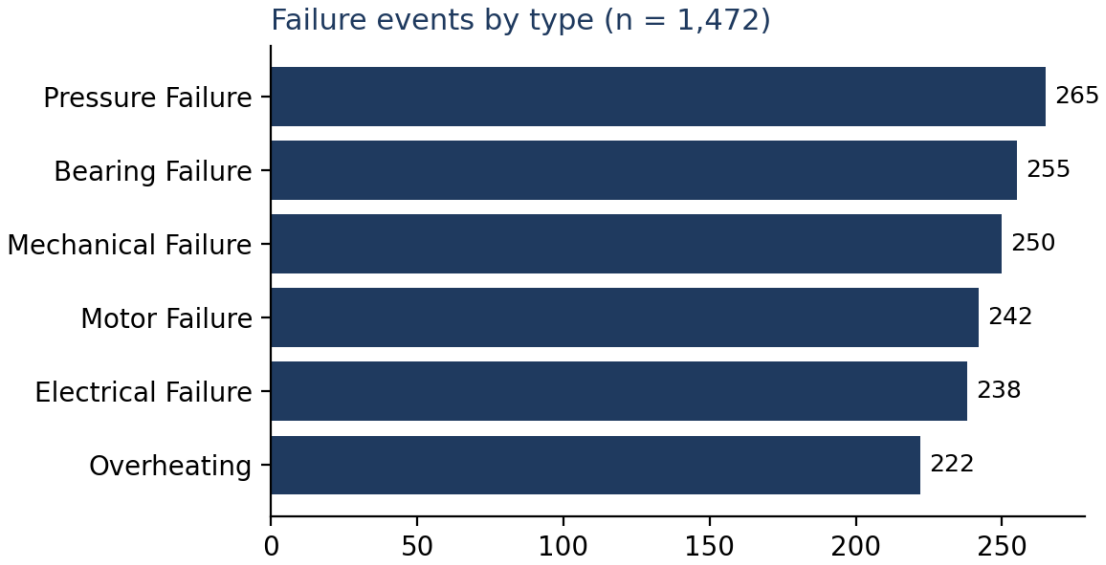


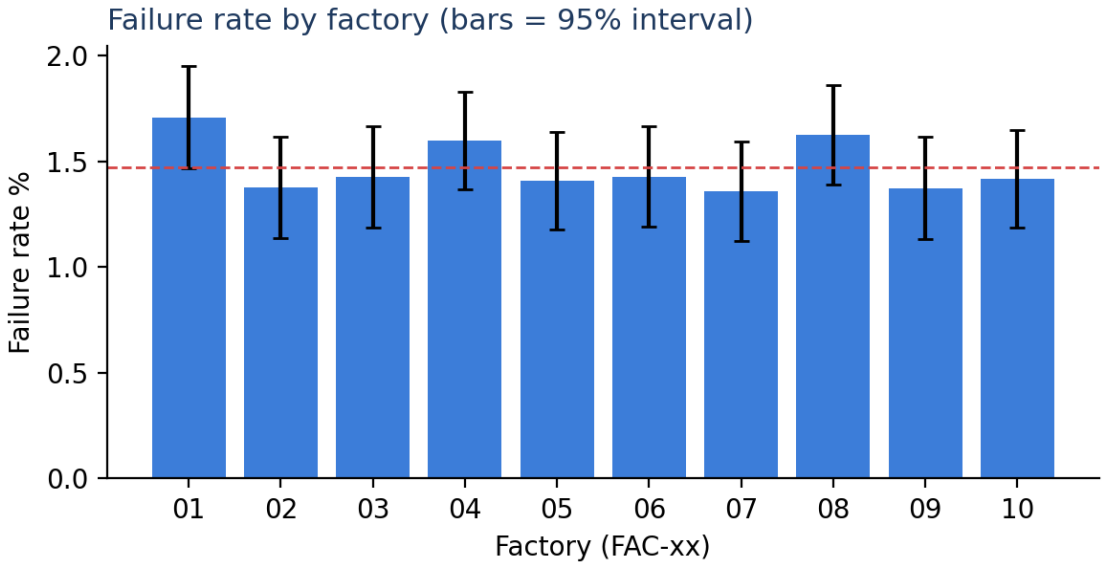




Wireframes - replace with final Power BI screenshots before presenting.

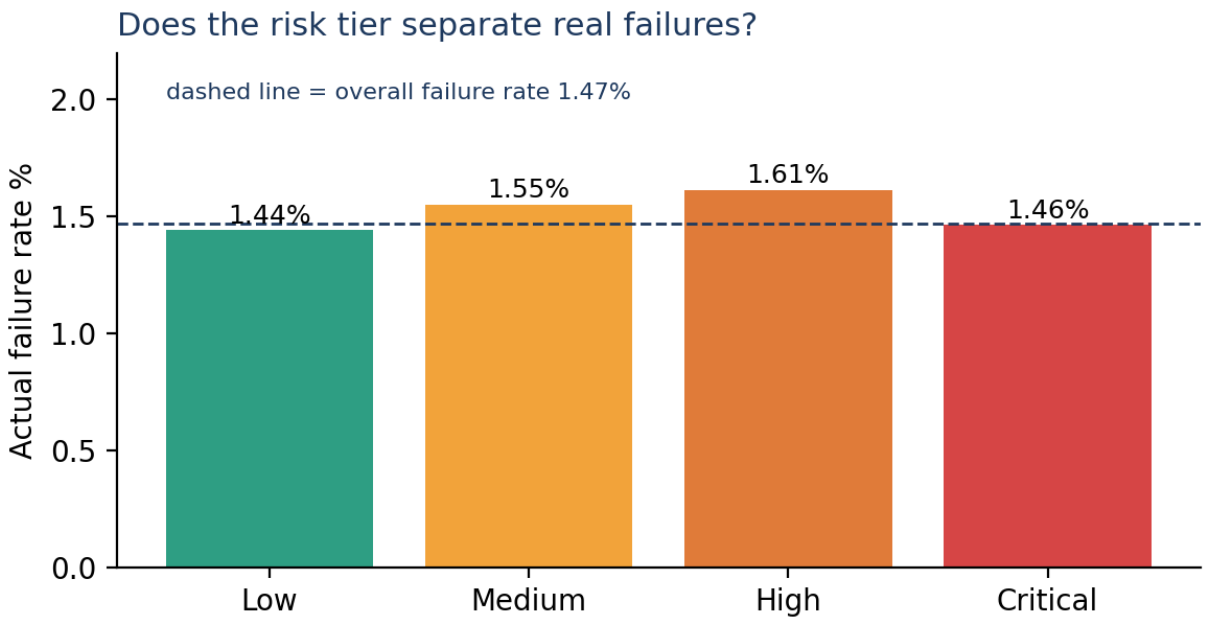
What the data shows





- 1,472 failure events in 100,000 readings; spread almost evenly across 6 types (222-265 each).
- Factory failure rates range 1.36%-1.71%, but a chi-square test gives p = 0.41: consistent with chance, so no factory ranking.

Validation: what the scores can and cannot do



- Derived risk tiers (Low/Medium/High/Critical) all show about the same actual failure rate (1.44%-1.61%).
- No column correlates with the failure label above ~0.007; a cross-validated model scored AUC 0.505 (0.5 = random).
- So: the dashboard is for monitoring and reporting - not a validated failure predictor.

A real predictive model would need data where failures are linked to sensor history.

Conclusions and next steps

- Delivered: reproducible ETL, SQL Server star schema, DAX measures, theme, 4-page dashboard design, documentation.
- Learned: always validate a dataset's claims before building on them.
- Next: add row-level drill-through, scheduled refresh, and test the model on data with real failure signals.

Thank you

Questions?

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